

## Question I

You are provided with a dataset of daily website traffic for an online retail store, including metrics like page views, unique visitors, bounce rate, and time spent on site over the last quarter. Choose two relevant metrics from the dataset and create visualizations (e.g., line charts, stacked area charts) to analyze the trends in website engagement over time. Identify any significant patterns or anomalies and discuss potential reasons behind them.

Metrics: Page Views, Unique Visitors, Bounce Rate, Time Spent

1. Line Chart – Page views trend
2. Line Chart – Unique visitors trend
3. Stacked Area Chart – Page views and unique visitors together
4. Area Chart – Total engagement trend
5. Column Chart – Comparison of daily page views
6. Bar Chart – Comparison of daily unique visitors
7. Scatter Plot – Relationship between time spent and bounce rate
8. Combo Chart – Page views (columns) and time spent (line)

You are given a dataset containing weekly mobile app usage statistics for a fitness tracking application. The dataset includes metrics such as active users, session duration, number of workouts logged, and user retention rate over the past six months.

Select any two relevant metrics from the dataset and create suitable visualizations (e.g., line charts, bar charts, or area charts) to analyze how user engagement has changed over time.

Identify any notable trends, sudden spikes or drops, and explain possible reasons for these patterns based on app usage behavior.

Metrics: Active Users, Session Duration, Workouts Logged, Retention Rate

1. Line Chart – Weekly active users trend
2. Line Chart – Weekly session duration trend
3. Area Chart – Overall usage pattern
4. Column Chart – Workouts logged per week
5. Bar Chart – Retention rate per week
6. Scatter Plot – Session duration versus retention rate
7. Stacked Column Chart – Active users and workouts together
8. Combo Chart – Active users (columns) and session duration (line)

You are provided with a dataset containing **monthly sales performance** for a chain of electronics stores, including metrics such as **total sales revenue, number of units sold, customer footfall, and average transaction value** over the past year.

Choose **two relevant metrics** from the dataset and create appropriate **visualizations** (e.g., line

graphs, bar charts, or area charts) to examine the trends in sales performance over time.

Identify any **significant trends, seasonal variations, or anomalies**, and discuss possible factors that might explain these observations.

Metrics: Sales Revenue, Units Sold, Footfall, Average Transaction Value

1. Line Chart – Monthly sales revenue trend
2. Line Chart – Monthly units sold trend
3. Area Chart – Footfall trend
4. Column Chart – Units sold per month
5. Bar Chart – Average transaction value per month
6. Scatter Plot – Footfall versus revenue
7. Stacked Column Chart – Revenue and units sold together
8. Combo Chart – Revenue (line) and units sold (columns)

You are given a dataset containing quarterly financial performance data for a chain of retail clothing stores, including metrics such as total revenue, number of transactions, average purchase value, and customer return rate over the past year.

Choose two relevant metrics from the dataset and create suitable visualizations (e.g., line charts, bar charts, or area charts) to analyze the changes in financial performance over time.

Identify any major trends, seasonal effects, or unusual spikes/drops, and explain possible reasons that could account for these patterns.

Metrics: Revenue, Transactions, Average Purchase Value, Return Rate

1. Line Chart – Quarterly revenue trend
2. Line Chart – Quarterly transactions trend
3. Area Chart – Average purchase value trend
4. Column Chart – Return rate per quarter
5. Bar Chart – Quarterly financial comparison
6. Scatter Plot – Purchase value versus return rate
7. Stacked Column Chart – Revenue and transactions
8. Combo Chart – Revenue (line) and return rate (columns)

## Question II

You have data from three different marketing campaigns, including the budget allocated, the number of leads generated, and the conversion rate for each campaign. Visualize the distribution of leads generated by each campaign using appropriate techniques (e.g., box plots, violin plots). Additionally, create a visualization to compare the cost per lead across the three campaigns. Based on your visualizations, which campaign appears to be the most efficient?

Metrics: Budget, Leads, Conversion Rate

1. Box Plot – Leads distribution
2. Violin Plot – Leads density
3. Column Chart – Leads per campaign
4. Bar Chart – Conversion rate comparison
5. Line Chart – Budget versus leads
6. Scatter Plot – Relationship between budget and leads
7. Area Chart – Cumulative leads trend
8. Combo Chart – Budget (columns) and leads (line)

You are provided with data from three different product launches, including the marketing spend, number of customer sign-ups, and retention rate for each launch. Visualize the distribution of customer sign-ups for each product launch using suitable plots (e.g., box plots, violin plots). Next, create a visualization to compare the cost per retained customer across the three product launches. Based on your visual analysis, identify which product launch demonstrates the highest efficiency.

Metrics: Marketing Spend, Sign-ups, Retention Rate

1. Box Plot – Sign-ups distribution
2. Violin Plot – Sign-ups density
3. Column Chart – Cost per sign-up
4. Bar Chart – Retention rate comparison
5. Line Chart – Marketing spend versus sign-ups
6. Scatter Plot – Sign-ups versus retention rate
7. Area Chart – Total sign-ups trend
8. Combo Chart – Spend (columns) and retained users (line)

You have data from four different customer support channels, including the average response time, number of queries handled, and customer satisfaction score for each channel. Visualize the distribution of response times across the support channels using appropriate techniques (e.g., box plots, violin plots). Additionally, create a visualization to compare the satisfaction score per query across the four channels. Based on your visualizations, determine which support channel appears to deliver the best overall performance.

Metrics: Response Time, Queries Handled, Satisfaction Score

1. Box Plot – Response time distribution
2. Violin Plot – Response time density
3. Column Chart – Queries handled per channel
4. Bar Chart – Satisfaction score across channels
5. Line Chart – Response time trend
6. Scatter Plot – Satisfaction versus response time
7. Area Chart – Overall query volume trend

### 8. Combo Chart – Queries (columns) and satisfaction (line)

You are provided with data from four advertising channels, including the advertising budget, number of leads generated, and the lead-to-customer conversion rate for each channel.

Visualize the distribution of leads generated across the four channels using suitable plots (e.g., box plots, violin plots). Then, create a visualization to compare the cost per converted customer across these advertising channels. Based on your visual analysis, determine which advertising channel appears to be the most cost-efficient.

Metrics: Budget, Leads, Conversion Rate

1. Box Plot – Leads distribution
2. Violin Plot – Leads density
3. Column Chart – Cost per converted customer
4. Bar Chart – Leads per channel
5. Line Chart – Conversion rate trend
6. Scatter Plot – Budget versus leads generated
7. Area Chart – Cumulative leads trend
8. Combo Chart – Budget (columns) and conversions (line)

### Question III

You have collected data from a class of students, recording the number of hours each student studied for a particular exam and their corresponding score. Formulate a null and alternative hypothesis regarding the relationship between study hours and exam scores. Perform a correlation analysis and, if appropriate, a simple linear regression to investigate this relationship. Interpret the statistical results to either support or reject your hypothesis at a chosen significance level.

```
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import pearsonr
from sklearn.linear_model import LinearRegression
df = pd.read_csv("study_data.csv")
X = df[['Study_Hours']]
Y = df['Exam_Score']
corr, p = pearsonr(df['Study_Hours'], df['Exam_Score'])
print("Correlation:", corr)
print("P-value:", p)
model = LinearRegression()
model.fit(X, Y)
print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)
print("R2 Score:", model.score(X, Y))
```

```
plt.scatter(df['Study_Hours'], df['Exam_Score'])
plt.plot(df['Study_Hours'], model.predict(X), color='red')
plt.xlabel("Study Hours")
plt.ylabel("Exam Score")
plt.title("Study Hours vs Exam Score")
plt.show()
```

You have collected data from a fitness study, recording the number of weekly exercise hours for each participant and their corresponding resting heart rate.

Formulate a null and alternative hypothesis about the relationship between weekly exercise hours and resting heart rate. Perform a correlation analysis and, if appropriate, conduct a simple linear regression to investigate this relationship. Interpret the statistical results and determine whether your hypothesis is supported or rejected at a chosen significance level.

```
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import pearsonr
from sklearn.linear_model import LinearRegression
df = pd.read_csv("fitness_data.csv")
X = df[['Exercise_Hours']]
Y = df['Resting_Heart_Rate']
corr, p = pearsonr(df['Exercise_Hours'], df['Resting_Heart_Rate'])
print("Correlation:", corr)
print("P-value:", p)
model = LinearRegression()
model.fit(X, Y)

print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)
print("R2 Score:", model.score(X, Y))
plt.scatter(df['Exercise_Hours'], df['Resting_Heart_Rate'])
plt.plot(df['Exercise_Hours'], model.predict(X), color='red')
plt.xlabel("Exercise Hours")
plt.ylabel("Resting Heart Rate")
plt.title("Exercise Hours vs Resting Heart Rate")
plt.show()
```

You have collected data from a sleep study, recording the average number of hours each participant sleeps per night and their corresponding daytime productivity score.

Formulate the null and alternative hypotheses regarding the relationship between sleep duration and productivity. Perform a correlation analysis and, if appropriate, conduct a simple linear regression to examine the strength and direction of this relationship.

Interpret the statistical results and conclude whether your hypothesis is supported or rejected at a chosen significance level.

```
import pandas as pd
```

```

import matplotlib.pyplot as plt
from scipy.stats import pearsonr
from sklearn.linear_model import LinearRegression
df = pd.read_csv("sleep_data.csv")
X = df[['Sleep_Hours']]
Y = df[['Productivity_Score']]
corr, p = pearsonr(df['Sleep_Hours'], df['Productivity_Score'])
print("Correlation:", corr)
print("P-value:", p)
model = LinearRegression()
model.fit(X, Y)
print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)
print("R2 Score:", model.score(X, Y))
plt.scatter(df['Sleep_Hours'], df['Productivity_Score'])
plt.plot(df['Sleep_Hours'], model.predict(X), color='red')
plt.xlabel("Sleep Hours")
plt.ylabel("Productivity Score")
plt.title("Sleep Hours vs Productivity Score")
plt.show()

```

You have collected data from a nutrition study, recording the daily calorie intake of each participant and their corresponding body weight. Formulate a null and alternative hypothesis regarding the relationship between daily calorie intake and body weight. Conduct a correlation analysis and, if appropriate, perform a simple linear regression to further assess the relationship. Interpret your statistical results and determine whether the hypothesis is supported or rejected at a selected significance level.

```

import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import pearsonr
from sklearn.linear_model import LinearRegression

df = pd.read_csv("nutrition_data.csv")
X = df[['Calorie_Intake']]
Y = df[['Body_Weight']]
corr, p = pearsonr(df['Calorie_Intake'], df['Body_Weight'])
print("Correlation:", corr)
print("P-value:", p)
model = LinearRegression()
model.fit(X, Y)
print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)
print("R2 Score:", model.score(X, Y))
plt.scatter(df['Calorie_Intake'], df['Body_Weight'])
plt.plot(df['Calorie_Intake'], model.predict(X), color='red')

```

```
plt.xlabel("Calorie Intake")
plt.ylabel("Body Weight")
plt.title("Calorie Intake vs Body Weight")
plt.show()
```

#### Question IV

You have a dataset containing information about customers of a subscription service, including their demographics, usage patterns, and whether they churned (cancelled their subscription) in the last month. Build a logistic regression model to predict the probability of customer churn based on the available features. Identify the key features that have the most significant impact on churn probability based on the model's coefficients.

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
df = pd.read_csv("customer_churn.csv")
y = df['Churn']
X = df.drop('Churn', axis=1)
X = pd.get_dummies(X, drop_first=True)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)
pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, pred))
coeff = pd.DataFrame({'Feature': X.columns, 'Coefficient': model.coef_[0]})
print(coeff.sort_values(by='Coefficient', ascending=False))
```

You are given a dataset containing information about patients in a hospital, including their demographic details, lifestyle habits, medical test results, and whether they were readmitted within 30 days. Build a logistic regression model to predict the probability of patient readmission based on the available features. Using the model's coefficients, identify which features have the strongest influence on the likelihood of readmission.

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

df = pd.read_csv("patient_readmission.csv")
y = df['Readmitted']
X = df.drop('Readmitted', axis=1)
X = pd.get_dummies(X, drop_first=True)
```

```

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)
pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, pred))
coeff = pd.DataFrame({'Feature': X.columns, 'Coefficient': model.coef_[0]})
print(coeff.sort_values(by='Coefficient', ascending=False))

```

You are provided with a dataset from an online learning platform containing information about learners, such as their demographic details, course activity logs, assignment completion rates, and whether they dropped out of the course before completion. Build a logistic regression model to predict the probability of learner dropout using the available features. Based on the model's coefficients, identify the most influential factors contributing to the likelihood of dropout.

```

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
df = pd.read_csv("learner_dropout.csv")
y = df['Dropped_Out']
X = df.drop('Dropped_Out', axis=1)
X = pd.get_dummies(X, drop_first=True)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)
pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, pred))
coeff = pd.DataFrame({'Feature': X.columns, 'Coefficient': model.coef_[0]})
print(coeff.sort_values(by='Coefficient', ascending=False))

```

You are provided with a dataset from an e-commerce platform containing customer information such as demographics, browsing history, purchase frequency, and whether the customer made a repeat purchase in the last quarter. Build a logistic regression model to predict the probability of a repeat purchase based on the available features. Using the model's coefficients, identify the key features that most strongly influence the likelihood of a customer making a repeat purchase.

```

import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
df = pd.read_csv("repeat_purchase.csv")
y = df['Repeat_Purchase']
X = df.drop('Repeat_Purchase', axis=1)

```



```
X = pd.get_dummies(X, drop_first=True)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)
pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, pred))
coeff = pd.DataFrame({'Feature': X.columns, 'Coefficient': model.coef_[0]})
print(coeff.sort_values(by='Coefficient', ascending=False))
```

## Question V

You are receiving a continuous stream of data from air quality sensors across a city, reporting levels of pollutants like PM2.5 and CO. The goal is to detect sudden spikes in pollution levels in real-time. Describe a conceptual approach for processing this streaming data to identify instances where the concentration of PM2.5 exceeds a predefined threshold within a sliding time window. Outline the data processing steps and suggest a visualization method to alert city officials about these events in real-time. How could this analysis contribute to SDG 13 (Climate Action) in the context of public health?

## Air Quality Monitoring (PM2.5 & CO)

### 1. Real-Time Line Chart

- Shows PM2.5 levels continuously.

### 2. Real-Time Line Chart (CO)

- Shows CO concentration.

### 3. Area Chart

- Highlights rising pollution intensity.

### 4. Scatter Chart

- Relationship between PM2.5 and CO.

### 5. Clustered Column Chart

- Hourly average pollution levels.

### 6. Gauge Chart

- Shows current PM2.5 level relative to threshold.

## 7. Card Visual (Alert Value)

- Displays last PM2.5 reading with color coding.

## 8. Map Visualization (Color-Coded)

- Shows PM2.5 levels across different sensor locations.

You are receiving a continuous stream of data from **water quality monitoring sensors** placed along a major river, reporting parameters such as **pH level and dissolved oxygen (DO)** every few seconds. The objective is to **detect sudden drops in water quality** in real time.

Describe a **conceptual approach** for processing this streaming data to identify instances where the **dissolved oxygen level falls below a critical threshold** within a **sliding time window**. Outline the **data processing steps**, including data ingestion, windowing, threshold evaluation, and event detection. Propose a **real-time visualization method** (e.g., alerts on a dashboard, color-coded map, or live time-series graph) to notify environmental authorities when such events occur. Finally, explain how this analysis could support **SDG 6 (Clean Water and Sanitation)** by improving environmental monitoring and protecting public health.

# Water Quality Monitoring (pH & Dissolved Oxygen)

## 1. Real-Time Line Chart (DO Levels)

- Detect sudden drops in DO.

## 2. Real-Time Line Chart (pH Levels)

- Shows fluctuations in water acidity.

## 3. Area Chart

- Highlights abnormal drops in DO.

## 4. Scatter Chart

- Relationship between pH and DO.

## 5. Clustered Column Chart

- Hourly changes in DO.

## 6. Gauge Chart

- Shows DO level vs minimum safe threshold.

#### **7. Card Visual (DO Alert Value)**

- Red/yellow/green indicator for DO safety zone.

#### **8. Map Visualization (River Locations)**

- Shows DO or pH levels along river sensors.